

Question 1A State Representation

[4 marks]

There are many possible solutions and credit will be award for any reasonable (and correct) solution. The most straightforward way to formulate this problem as uninformed search would be list all the variables (letters) in the puzzle. Let these variables be v_i , for $i = 1, \dots, m$, $m \leq 10$.

The state of the search problem is a mapping of $(v_1, \dots, v_m)$ to digits. Some of the mappings could be null.

[+2] for the correct state representation.

[+2] for stating the correct actions with corresponding state transitions.

Question 1B Invariant

[2 marks]

$v_i \neq v_j$, for all i, j where $i \neq j$ and v_i, v_j not null.

Question 1C Initial and goal states

[4 marks]

Initial state: $(null, null, \dots, null)$

Goal state: We won't know the goal state. If we knew the goal state, the problem is already solved! What we have is a goal test. The goal test has 2 parts:

1. There is a complete mapping, i.e., $v_i \neq null$, for all i
2. The addition relationship is valid.

[+2] for initial state.

[+2] for goal test.

Question 1D Which of the following statement(s) is/are true?

[4 marks]

- ☒ The search tree is finite.
- ☐ There are possibly many repeated states.
- ☒ There are possibly multiple goal states.
- ☐ We can always find an answer (valid mapping) if we employ the right search algorithm.
- ☐ None of the above.

Question 1E Best search algorithm.

[4 marks]

DFS, because it works and it does so with constant space.
Not BFS, because while it works, it takes more space than DFS.
Not DLS, because we don't know if we can find the solution if we limit the depth.
Not IDS, because while it works, We are paying extra for no good reason.

[+1] each for brief explanations for each algorithm.

Question 1F To Graph-Search or not? That is the question.

[2 marks]

No, there is no savings since we don't have repeated states. This question is here to test that the students understand the difference between TREE-SEARCH and GRAPH-SEARCH.

Question 1G Order of Growth (Time)

[2 marks]

This is somewhat of a trick question. The search tree is bounded by $O(10^m)$. In fact, we know that there are at most $10!$ possibilities.

Question 1H Make it run faster!

[3 marks]

Recall that a key technique to improve search is *pruning*. In this case, what we can do is to arrange to choose the letters at the lower decimal places to be used as the nodes earlier in the search. Then we also add an addition invariant/constraint that the first J columns from the right are all valid, i.e., when we added up the columns, there are no conflicts.

To make this more concrete, for the SEND-MORE-MONEY problem, we will choose letters for the nodes in the following order $\{D, E, Y, N, R, O, S, M\}$.

Question 1I Initial candidate solution.

[2 marks]

There are a several ways to formulate this problem. The candidate solution needs only be a plausible solution to the problem.

For each v_i , assign at random one of the digits from 0 to 9, while ensuring that $v_i \neq v_j$, for all $i \neq j$ (i.e., that there are no repeated digits).

Question 1J Heuristic function.

[4 marks]

There are 2 obvious candidates for heuristics:

1. Let $h = 0$ be the heuristic variable. Consider each of the $n + 1$ columns from right to left. $h++$ if the column does not add up. We take carry overs into account since we are considering the columns from right to left.
2. Maintain a counter for each letter. Consider each of the $n + 1$ columns from right to left. If the column does not add up, add one to the counters for all the letters in the column. We take carry overs into account since we are considering the columns from right to left. The heuristic is the sum of all the counters.

We are counting conflicts, so if the heuristic is zero, we have our answer.

[+3] for a valid heuristic described clearly.

[+1] for highlighting that heuristic = 0 is a goal test.

Question 1K Transition functionn.

[4 marks]

There are 2 cases:

1. If all the digits are used, i.e., $m = 10$. Then we just pick a random pair of letters that are involved in some conflict and swap them if we can reduce the heuristic.
2. If not all the digits are used, then with some probability p , we swap out the letter that is most conflicted with one of the unused digits.

There are many possible solutions for Parts (H), (I) and (J), but they need to be coordinated.

[+4] for reasonable transition based on Parts (H) and (I).

[-2] if some transitions are missing, i.e., if $m < 10$ and there is no operation to allow the unused digit to be swapped in.

Question 1L Which is better: informed search or local search?

[3 marks]

The answer is that it depends. If n is large, then it is likely that there are many constraints that will allow efficient pruning, so informed search is better. If n is small and there are many possible solutions, then local search would likely be very fast.

[+1] for it depends.

[+2] for explaining “it depends” correctly.

Question 2A Is h_1 admissible?

[2 marks]

Tests that student understands that an admissible heuristic for a relaxed problem must also be admissible for a more constrained problem.

[2m] Yes it is admissible, since this is the exact solution to the related problem where the coin can be directly moved to its goal position.

Question 2B Is h_2 admissible?

[2 marks]

This question tests that the student understands if an admissible heuristic must not overestimate the actual distance to the goal.

[2m] Yes it is admissible, since this never overestimates the distance of moving a coin to its goal position. It is a stricter version of the square grid since (1,3) has to take an actual path of 3 time steps to reach (2,3).

Question 2C Maximum Branching Factor

[2 marks]

$3k$, since each of the k open spaces can possibly accept a maximum of 3 different tiles.

[2m] state $3k$ with explanation

[1m] correct observations but incorrect final answer, or only state.

[0m] incorrect answers without explanation

Question 2D h_A admissible and consistent? Explain.

[4 marks]

☐ Admissible ☒ Not admissible

h_A is not admissible because suppose we have one coin adjacent to an open goal position, and another coin at the other (far) end of the puzzle, also adjacent to an open goal position.

c...c

g...g

Then the heuristic will overestimate the true value 2.

☐ Consistent ☒ Not consistent

h_A is not consistent because it is possible for the heuristic to decrease by more than 1 (the actual cost) if a coin moves to its adjacent open goal position as in the scenario given above.

Question 2E h_B admissible and consistent? Explain.

[4 marks]

☒ Admissible ☐ Not admissible

h_B is admissible because the actual distance to the goal state must at least minimally be solved by moving all misplaced coins to their closest goal locations.

☒ Consistent ☐ Not consistent

h_B is consistent because the heuristic decreases by no more than 1 (the actual cost) with each time step.

Question 2F h_C admissible and consistent? Explain.

[4 marks]

☐ Admissible ☒ Not admissible

h_C is inadmissible because this sums the distances of each coin to the most distant goal position; but the optimal solution might require the coin to be placed at a closer goal position instead, causing this heuristic to overestimate.

☐ Consistent ☒ Not consistent

h_C is inconsistent because the heuristic decreases by more than 1 (the actual cost) in a time step, if a coin is placed on a far-away goal position.

Question 2G Dominant relationships between the heuristics.

[5 marks]

h_C is clearly dominant over h_B since it takes the maximum distance.

h_C is also dominant over h_A since h_A only takes the minimum of all the maximum distances, while h_C takes the sum of the maximum distances.

[+1] $h_C > h_B$

[+1] $h_C > h_A$

[+1] No dominant relationship between h_A and h_B

Between h_B and h_A , there is no dominant relationship since it is possible for h_B to sometimes be greater than h_A when C is all near the top of the puzzle, and G is all near the bottom, and sometimes h_A to sometimes be greater than h_B when there are only two coins as in the configuration described in 2D's answer.

Question 3A To save or to spend?

[2 marks]

Ben should save the coin.

If the coin is small, the decision tree shows that Ben should save.

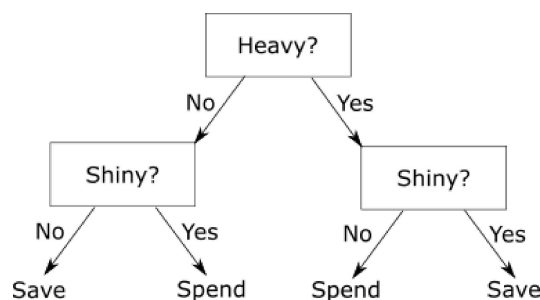
If the coin is big, then it is also heavy and shiny, hence should save too.

This question tests that the student knows how to reason about a decision tree.

Question 3B Minimal-Depth Tree.

[2 marks]

This question is here to test that the student knows how to reconfigure a decision tree.



Question 3C What is the entropy?

[2 marks]

The counts of the 10 decisions are 5 for Save and 5 for Spend.

Since there are two classes of equal distribution, the entropy can simply be observed to be 1.

$$\begin{aligned}\text{Entropy} &= -\frac{5}{10} \log\left(\frac{5}{10}\right) - \frac{5}{10} \log\left(\frac{5}{10}\right) \\ &= +0.5 + 0.5 = 1.00\end{aligned}$$

Question 3D One-level Tree.

[6 marks]

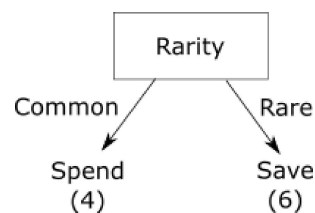
We only need to consider the smallest remaining entropy. In this case, we will use Rarity of the coin to split.

$$\begin{aligned}\text{Remainder(Rarity)} &= \frac{6}{10} I\left(\frac{1}{6}, \frac{5}{6}\right) + \frac{4}{10} I\left(\frac{4}{4}, 0\right) \\ &= -0.6\left(\frac{1}{6} \log_2\left(\frac{1}{6}\right) + \frac{5}{6} \log_2\left(\frac{5}{6}\right)\right) - 0 \\ &= 0.390 \approx 0.39\end{aligned}$$

$$\begin{aligned}\text{Remainder(Vintage)} &= \frac{5}{10} I\left(\frac{3}{5}, \frac{2}{5}\right) + \frac{5}{10} I\left(\frac{2}{5}, \frac{3}{5}\right) \\ &= -0.5\left(\frac{3}{5} \log_2\left(\frac{3}{5}\right) + \frac{2}{5} \log_2\left(\frac{2}{5}\right)\right) \times 2 \\ &= 0.970 \approx 0.97\end{aligned}$$

$$\begin{aligned}\text{Remainder(Condition)} &= \frac{7}{10} I\left(\frac{4}{7}, \frac{3}{7}\right) + \frac{3}{10} I\left(\frac{1}{3}, \frac{2}{3}\right) \\ &= -0.7\left(\frac{4}{7} \log_2\left(\frac{4}{7}\right) + \frac{3}{7} \log_2\left(\frac{3}{7}\right)\right) - 0.3\left(\frac{1}{3} \log_2\left(\frac{1}{3}\right) + \frac{2}{3} \log_2\left(\frac{2}{3}\right)\right) \\ &= 0.965 \approx 0.97\end{aligned}$$

$$\begin{aligned}\text{Remainder(Production)} &= \frac{2}{10} I\left(\frac{1}{2}, \frac{1}{2}\right) + \frac{8}{10} I\left(\frac{4}{8}, \frac{4}{8}\right) \\ &= 1.00\end{aligned}$$



Question 3E Full Decision Tree.

[5 marks]

Using the results from part (D), we split on rarity, so we now only consider “Rare” coins.

Coin	Is Vintage	Condition	In Production	Decision
1	No	Rusty	Inactive	Spend
2	No	Rusty	Active	Save
3	Yes	Rusty	Active	Save
4	Yes	Rusty	Active	Save
9	No	Mint	Active	Save
10	Yes	Mint	Inactive	Save

Again, consider the smallest remaining entropy. In this case, we will use Production next.

$$\begin{aligned}
 \text{Remainder(Vintage)} &= \frac{3}{6}I\left(\frac{2}{3}, \frac{1}{3}\right) + \frac{3}{6}I\left(\frac{3}{3}, 0\right) \\
 &= -0.5\left(\frac{2}{3}\log_2\left(\frac{2}{3}\right) + \frac{1}{3}\log_2\left(\frac{1}{3}\right)\right) \\
 &= 0.459 \approx 0.46
 \end{aligned}$$

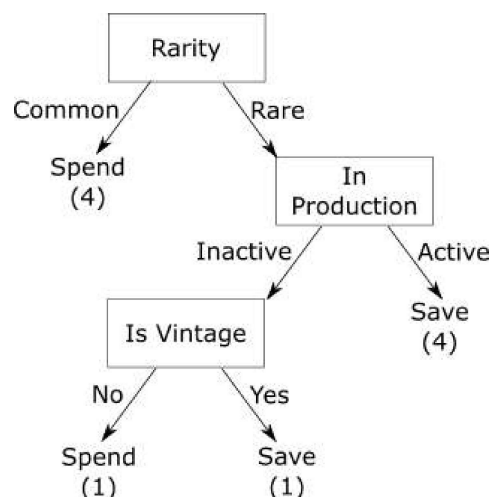
$$\begin{aligned}
 \text{Remainder(Condition)} &= \frac{4}{6}I\left(\frac{3}{4}, \frac{1}{4}\right) + \frac{2}{6}I\left(\frac{2}{2}, 0\right) \\
 &= -\frac{2}{3}\left(\frac{3}{4}\log_2\left(\frac{3}{4}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right)\right) \\
 &= 0.540 \approx 0.54
 \end{aligned}$$

$$\begin{aligned}
 \text{Remainder(Production)} &= \frac{2}{6}I\left(\frac{1}{2}, \frac{1}{2}\right) + \frac{4}{6}I\left(\frac{4}{4}, 0\right) \\
 &= 0.333 \approx 0.33
 \end{aligned}$$

And are now left with only “Rare and Inactive” coins.

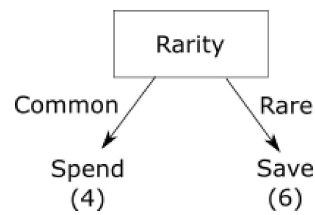
Coin	Is Vintage	Condition	Decision
1	No	Rusty	Spend
10	Yes	Mint	Save

Both provide the same information gain, so we tie-break, and use Vintage.



Question 3F Min-Sample DT

[4 marks]

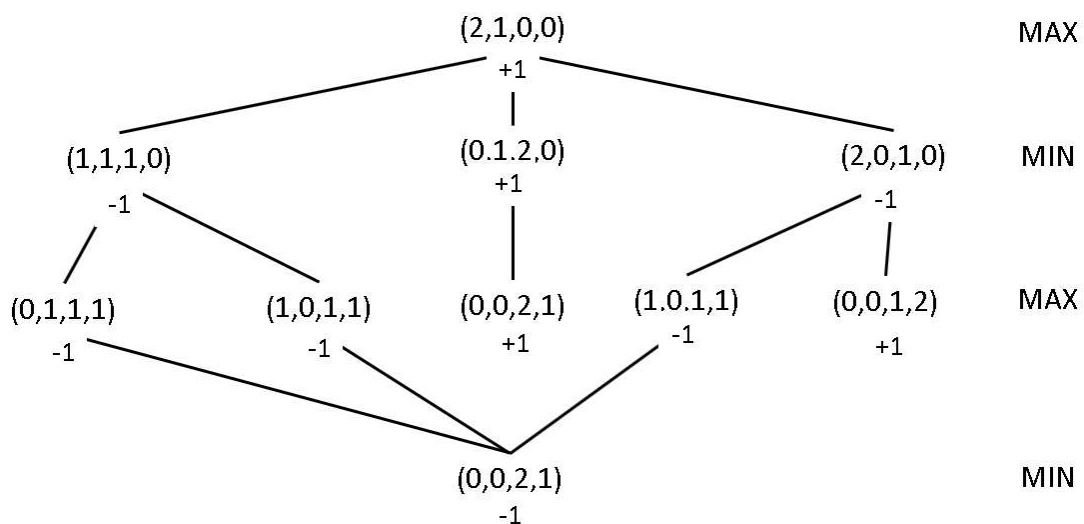


Coin 1 is likely an outlier.

Question 4A Nim Game Tree.

[6 marks]

This question is here to test that the student knows how to draw a game tree and solve it correctly. This question is meant to be easy. First player always wins.



[+3] for the correct tree and state representation.

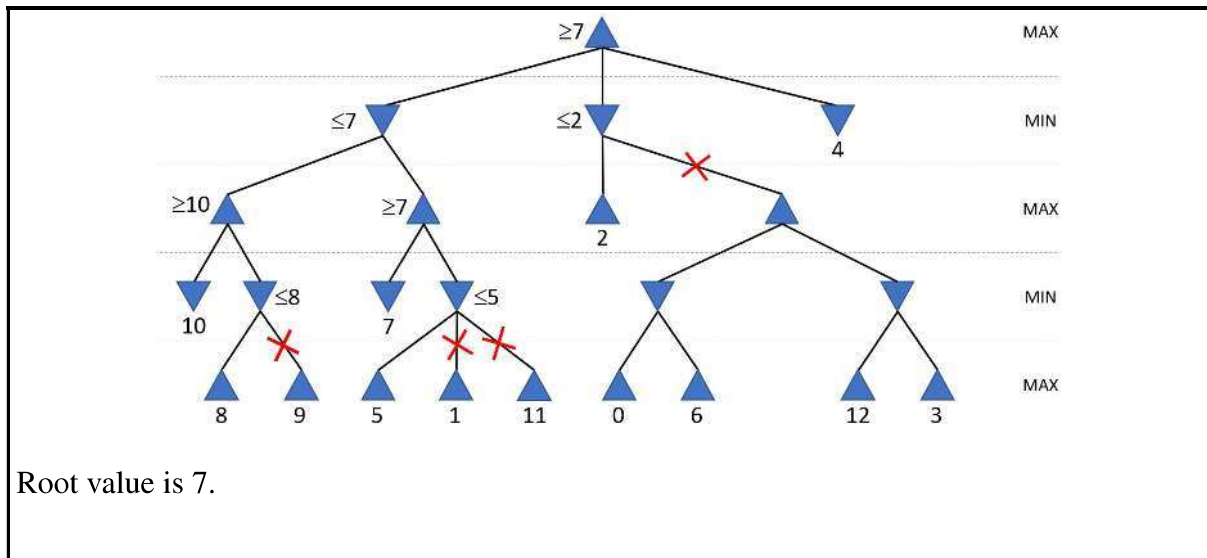
[+1] for stating that first player wins.

[+2] for solving the game tree correctly.

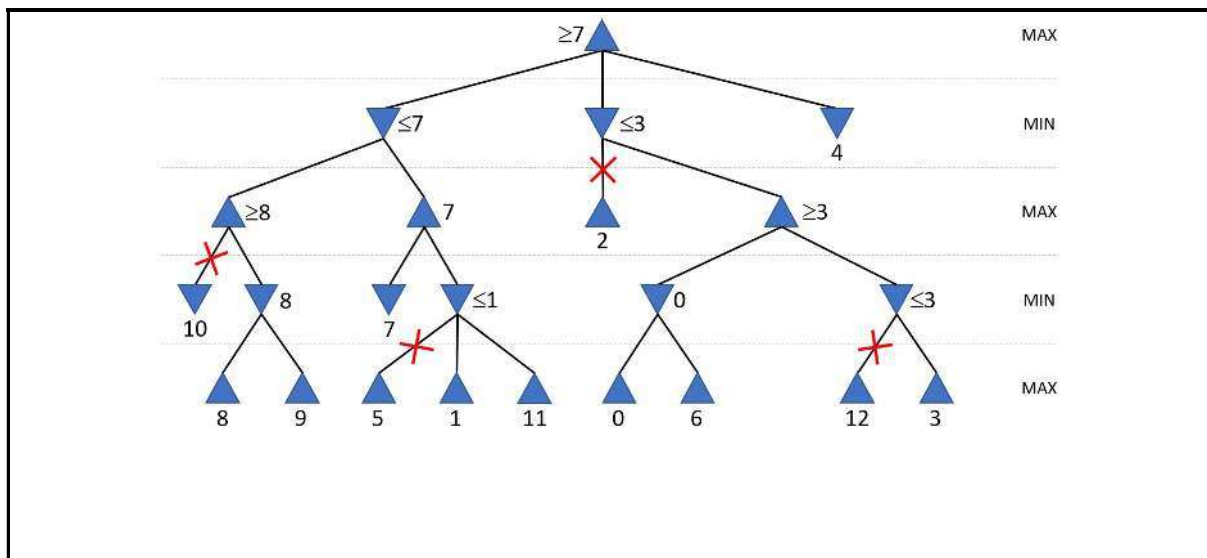
[-2] for each error.

Question 4B Alpha-beta left-to-right.

[5 marks]

**Question 4C** Alpha-beta right-to-left.

[4 marks]

**Question 5** 3 most important lessons

[3 marks]

There are no right answers here. Students will get credit for 3 well-explained learning points that are reasonable and justified. Student needs to put in *some* effort. Clearly if the student makes a patently false statement, marks will be deducted.